

Simple Games

- Games: a mathematical formalism for rational interaction
- What is the most popular solution concept? (Nash Equilibrium)

Types of Uncertainty

Alleatory



Markov Decision Process

Epistemic (Static)



Reinforcement Learning

Epistemic (Dynamic)



POMDP

Interaction



Game

Normal Form Games

- Alice and Bob are working on a homework assignment.
- They can either **share** or **withhold** their knowledge.
- If one player shares knowledge, the other benefits greatly, but the sharer also benefits by getting to test their knowledge

		Alice's Payoffs				Bob's Payoffs	
		B				B	
A		S	W	A		S	W
	S	4	2		S	4	3
	W	3	1		W	2	1

		Bob	
		S	W
Alice	S	4, 4	2, 3
	W	3, 2	1, 1

Called a **Normal Form, Simple**, or **Bimatrix** Game

Question for today: What **solution concept** should we use for games?

Dominant Strategies

		Bob	
		S	W
Alice	S	4, 4	2, 3
	W	3, 2	1, 1

- **Dominant (Pure) Strategy:** Action a is a dominant strategy if it is a best response to every action taken by the other player.
- **Dominant Strategy Equilibrium:** Every player plays a dominant strategy

Definitions

- Action $a^i \in A^i$
- Joint Action $a = (a^1, \dots, a^k)$
- All Other Actions
 $a^{-i} = (a^1, \dots, a^{i-1}, a^{i+1}, \dots, a^k)$
- Reward $R^i(a)$
- Joint Reward $R(a) = (R^1(a), \dots, R^k(a))$

Deterministic Best Response:

Action a^i is a deterministic best response to a^{-i} if

$$R^i(a^i, a^{-i}) \geq R^i(a^{i'}, a^{-i}) \quad \forall a^{i'}$$

Is the dominant strategy equilibrium always the best outcome for the players?

A more surprising example:

The Prisoner's Dilemma

- 2 criminals are captured
- Each can either keep silent (**S**) or testify (**T**)
 - other keeps silent → minor conviction (1 year)
 - other testifies → major conviction: 4 years
 - testify → 1 year removed from sentence

Player 1

Player 2

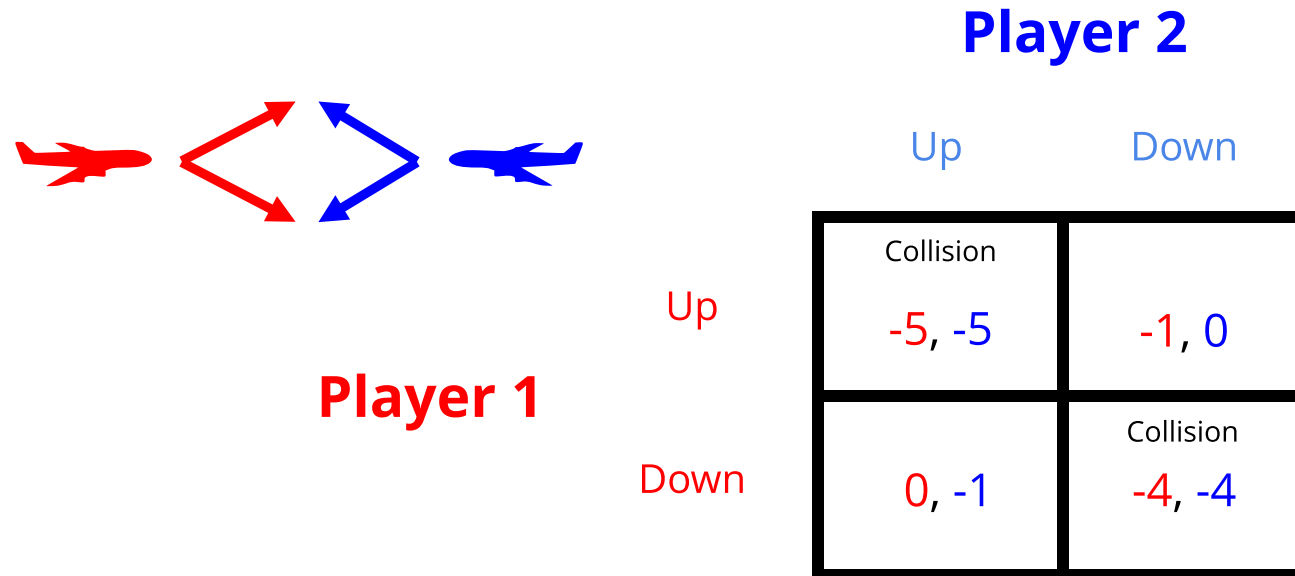
	S	T
S	-1, -1	-4, 0
T	0, -4	-3, -3

- Dominant strategy for both players is to testify
- Dominant strategy equilibrium is a very bad social result (for the criminals)

Do all simple games have a dominant strategy equilibrium?

Collision Avoidance Game

Example: Airborne Collision Avoidance



Pure Nash Equilibrium: All players play a deterministic best response.

Which equilibrium is better?

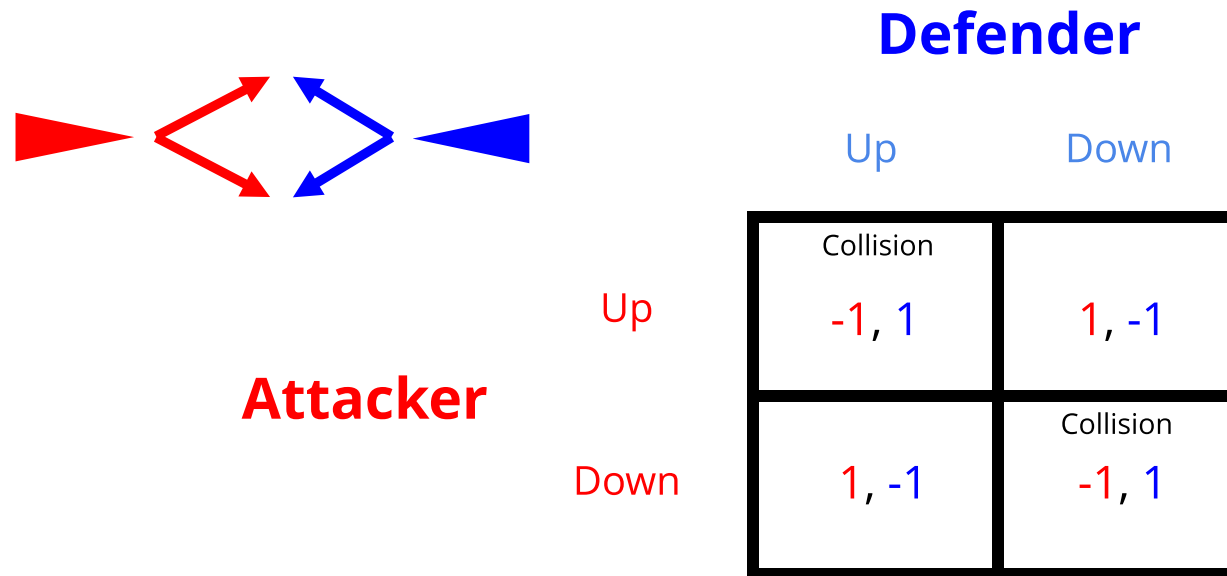
Do all simple games have a pure Nash equilibrium?

Practice: Find Pure Nash Equilibria

		Player 2		
		a	b	c
Player 1	a	4,4	2,5	0,0
	b	5,2	3,3	0,0
	c	0,0	0,0	10,10

Missile Defense Game

Missile Defense (simplified)



No Pure Nash Equilibrium!

Need a broader solution concept: Mixed Nash equilibrium.

Vocabulary and Notation for Mixed Strategies

	Single Player	Joint
• Action	$a^i \in A^i$	$a \in A$
• Policy (strategy)	$\pi^i(a^i)$	$\pi(a) = \prod_i \pi^i(a^i)$
• Reward	$R^i(a)$	$R(a)$
• Utility	$U^i(\pi) = \sum_a R^i(a)\pi(a)$	$U(\pi) = \sum_a R(a)\pi(a)$

Two Player Zero Sum:

$$R^1(a) + R^2(a) = 0 \quad \forall a$$

Cooperative:

$$R^i(a) = R^j(a) \quad \forall i, j, a$$

("General sum" used to indicate that the game is not limited to zero-sum or coop.)

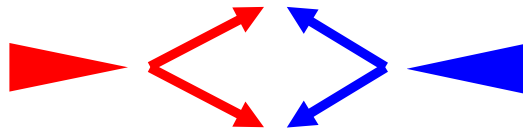
Best Response: Given a joint policy of all other agents, π^{-i} , a best response is a policy π^i that satisfies

$$U^i(\pi^i, \pi^{-i}) \geq U^i(\pi^{i'}, \pi^{-i})$$

for all other $\pi^{i'}$.

Missile Defense Game

Missile Defense (simplified)



Attacker

Up

Down

Defender

Up

Down

Collision -1, 1	1, -1
1, -1	Collision -1, 1

- A ***Nash equilibrium*** is a joint policy in which **all agents play a best response**

Rock-paper scissors

1. Guess the Nash Equilibrium argument
2. Make a qualitative argument that this is an NE based on best responses

		agent 2		
		rock	paper	scissors
agent 1	rock	0,0	-1,1	1,-1
	paper	1,-1	0,0	-1,1
	scissors	-1,1	1,-1	0,0

Do all simple games have at least one Nash equilibrium?

Yes!! (might be mixed)

Every finite game has a Nash Equilibrium

Kakutani's fixed-point theorem

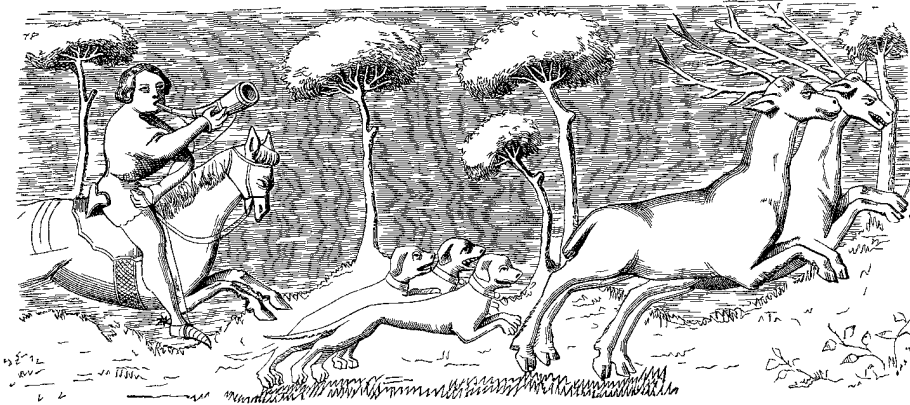
A correspondence $f: X \rightarrow X$ has a fixed point (i.e., $\mathbf{x} \in f(\mathbf{x})$ for some $\mathbf{x} \in X$) if all of the following conditions hold.

- (1) X is a non-empty, closed, bounded, and convex set.
- (2) $f(\mathbf{x})$ is non-empty for any $\mathbf{x}$.
- (3) $f(\mathbf{x})$ is convex for any $\mathbf{x}$.
- (4) The set $\{ (\mathbf{x}, \mathbf{y}) \mid \mathbf{y} \in f(\mathbf{x}) \}$ is closed.

- Let x be a joint strategy, π .
- Let f be BR , that is, the best response operator
- A fixed point of BR is a Nash Equilibrium
- The BR operator and policy space for finite games meet the conditions above
- BR has a fixed point for every finite game, i.e. every finite game has a Nash Equilibrium

Not on Exam

Calculating Mixed Nash



	Stag	Hare
Stag	4, 4	1, 3
Hare	3, 1	2, 2

- In a Mixed Nash Equilibrium, players must be *indifferent* between two or more actions
- (In large games, finding the support of the mixed strategies is the hard part)

General approach to find Nash Equilibria

$$\begin{aligned}
 &\underset{\pi, U}{\text{minimize}} && \sum_i (U^i - U^i(\pi)) \\
 &\text{subject to} && U^i \geq U^i(a^i, \pi^{-i}) \text{ for all } i, a^i \\
 &&& \sum_{a^i} \pi^i(a^i) = 1 \text{ for all } i \\
 &&& \pi^i(a^i) \geq 0 \text{ for all } i, a^i
 \end{aligned}$$

Not on Exam

Topology of bimatrix games:

4	1	6	5	4	3	2	1	6	5	4	3	2	1
2	3	4	2	3	4	2	3	4	2	3	4	2	3
1	4	2	1	4	2	1	4	2	1	4	2	1	4
Heg. Stability	Samaritan ₁	Samaritan ₂	Clock ₁	Clock ₂	Endless	Called Bluff	Bully	Unfair	Skewed BoS	Asym BoS	Chicken	—	1
3	3	4	2	3	4	2	3	4	2	3	4	2	3
1	4	2	1	4	2	1	4	2	1	4	2	1	4
Samson	Asym Sd ₁	Asym Sd ₂	Cycle ₁	Cycle ₂	Inspector	Self-serving ₁	Protector ₁	Protector ₂	Favorites ₁	Battle of Sexes	Asym BoS	—	1
4	3	4	2	3	4	2	3	4	2	3	4	2	3
2	1	4	2	1	4	2	1	4	2	1	4	2	1
Dellah	Asym Sd ₁	Asym Sd ₂	Pursuit	Pareto	Missile Crisis	Self-serving ₂	Protector ₁	Protector ₂	Hero	Favorites ₂	Skewed BoS	X	—
2	3	4	2	3	4	2	3	4	2	3	4	2	3
3	1	4	2	1	4	2	3	4	2	1	4	2	3
Hostage	Benevolence ₁	Benevolence ₂	2nd Best ₁	2nd Best ₂	Big Bully	Tragedy	Delight ₁	Pure Delight	Protector ₁	Protector ₂	Unfair	—	1
1	4	2	1	4	2	1	4	2	1	4	2	1	4
Blackmailer	Benevolence ₁	Benevolence ₂	2nd Best ₁	2nd Best ₂	Hamlet	Total Conflict	Mixed Delight	Delight ₂	Protector ₁	Protector ₂	Bully	—	1
1	4	2	1	4	2	1	4	2	1	4	2	1	4
Id. Hegemony	Samaritan ₁	Samaritan ₂	Revelation	Alibi	Asym Pd	Prisoners D.	Total Conflict	Tragedy	Self-serving ₁	Self-serving ₂	Called Bluff	—	1
2	3	4	2	3	4	2	3	4	2	3	4	2	3
1	4	2	1	4	2	1	4	2	1	4	2	1	4
Win-win'	C. Aligned ₁	C. Aligned ₂	C. Assurance ₁	C. Assurance ₂	Stag Hunt	Asym Pd'	Hamlet'	Big Bully'	Missile Crisis'	Inspector'	Endless'	—	1
3	3	4	2	3	4	2	3	4	2	3	4	2	3
1	4	2	1	4	2	1	4	2	1	4	2	1	4
R Assurance ₁	Commons ₁	Commons ₂	Coordination ₁	Coordination ₂	R Assurance ₁	Alibi'	2nd Best ₁	2nd Best ₂	Pareto'	Cycle ₁	Clock ₁	—	1
2	3	4	2	3	4	2	3	4	2	3	4	2	3
1	4	2	1	4	2	1	4	2	1	4	2	1	4
R Assurance ₂	Commons ₁	Commons ₂	Coordination ₁	Coordination ₂	R Assurance ₂	Revelation'	2nd Best ₁	2nd Best ₂	Pursuit'	Cycle ₂	Clock ₂	—	1
3	3	4	2	3	4	2	3	4	2	3	4	2	3
1	4	2	1	4	2	1	4	2	1	4	2	1	4
Row Aligned ₁	Harmony ₁	Harmony-mix	Commons ₁	Commons ₂	Row Aligned ₁	Samaritan ₁	Benevolent ₁	Benevolent ₂	Asym Sd ₁	Asym Sd ₂	Samaritan ₂	—	1
2	3	4	2	3	4	2	3	4	2	3	4	2	3
1	4	2	1	4	2	1	4	2	1	4	2	1	4
Row Aligned ₂	Harmony ₂	Harmony-pure	Commons ₁	Commons ₂	Row Aligned ₂	Samaritan ₂	Benevolent ₁	Benevolent ₂	Asym Sd ₁	Asym Sd ₂	Active Samr ₁	—	1
3	3	4	2	3	4	2	3	4	2	3	4	2	3
1	4	2	1	4	2	1	4	2	1	4	2	1	4
No Conflict	C. Aligned ₁	C. Aligned ₂	C. Assurance ₁	C. Assurance ₂	Win-win	Id. Hegemony	Blackmailer'	Hostage'	Dellah'	Samson'	Heg. Stability'	—	1
3	3	4	2	3	4	2	3	4	2	3	4	2	3
1	4	2	1	4	2	1	4	2	1	4	2	1	4

Algorithms that use best response

Iterated Best Response: randomly cycle between agents who play the best response for the current policy (converges to Nash for certain narrow classes of games)

Fictitious Play:

1. Estimate maximum likelihood policies for opponents:

$$\pi^j(a^j) \propto N(j, a^j)$$

2. Play best response to estimated policy

(The joint policy $\pi^j(a^j) \propto N(j, a^j) \quad \forall j, a^j$ converges to Nash for wide class of games, notably zero-sum)

Bach or Stravinsky

- Two people want to go to a concert
- P1 prefers Bach, P2 Stravinsky

	B	S
B	2, 1	0, 0
S	0, 0	1, 2

Correlated Equilibrium

- A *correlated joint policy* is a single distribution over the joint actions of all agents.
- A *correlated equilibrium* is a correlated joint policy where no agent i can increase their expected utility by deviating from the current policy.
- Easier to find than Nash equilibrium (Linear Program)

[https://youtube.com/shorts/w3q77ZZIqwA
?si=J8H6L6W5kTRs-mUx](https://youtube.com/shorts/w3q77ZZIqwA?si=J8H6L6W5kTRs-mUx)

Recap

- Games provide a mathematical framework for analyzing interaction between rational agents
- Games may not have a single "optimal" solution; instead there are equilibria that may not be rankable
- If every player is playing a best response, that joint policy is a Nash Equilibrium
- Every finite game has at least one Nash Equilibrium (pure or mixed)
- Mixed Nash equilibria occur when players are indifferent between two outcomes